```
r_arr = np.random.randint(1,51,size=10)
y = r_arr[:5]
z = r_arr[-3:]
print(r_arr)
print(y)
print(z)
```
✓ [35] 14ms
```
[ 8 46  6 43 23 36 14 22  6 48]
[ 8 46  6 43 23]
[22  6 48]
```

```
arr = np.array([[1,2,3],[4,5,6]])
SUM = arr.sum()
d_SUM = arr.sum(axis=0)
mean = arr.mean()
print(arr)
print(SUM)
print(d_SUM)
print(mean)
```
✓ [36] 15ms
```
[[1 2 3]
 [4 5 6]]
21
[5 7 9]
3.5
```

```
arr = np.random.randint(1, 100, size=10)
print(arr)
s_arr = np.sort(arr)
print(s_arr)
```
✓ [37] 14ms
```
[76 44 77 22 70 46 93 12 81 27]
[12 22 27 44 46 70 76 77 81 93]
```

```
arr = np.array([[1,2,3],[4,5,6]])
print(arr.shape)
print(arr.ndim)
print(arr.size)
print(arr.dtype)
```
✓ [32] 16ms
```
(2, 3)
2
6
int64
```

```
ar = np.arange(1,11)
print(ar)
```
✓ [33] 14ms
```
[ 1  2  3  4  5  6  7  8  9 10]
```

```
arr = ar.reshape(10, 1)
print(arr)
```
✓ [34] 14ms
```
[[ 1]
 [ 2]
 [ 3]
 [ 4]
 [ 5]
 [ 6]
 [ 7]
 [ 8]
 [ 9]
 [10]]
```

```
import numpy as np
import random
```
✓ [27] < 10 ms

```
a = np.array([1, 2, 3, 4])
print(a)
```
✓ [28] 19ms
```
[1 2 3 4]
```

```
b = np.array([[1, 2, 3],[5, 6, 7]])
print(b)
```
✓ [29] 15ms
```
[[1 2 3]
 [5 6 7]]
```

```
c = np.zeros((2,3))
print(c)
d = np.ones((2,3))
print(d)
```
✓ [30] 13ms
```
[[0. 0. 0.]
 [0. 0. 0.]]
[[1. 1. 1.]
 [1. 1. 1.]]
```

```
x = np.arange(0,12,2)
print(x)
```
✓ [31] 19ms
```
[ 0  2  4  6  8 10]
```

```
import numpy as np
import random
import pandas as pd
```
✓ [33] 14ms

```
matrix = np.array([[1,2,3],[4,5,6]])
print(matrix)
print(matrix[0,1])
print(matrix[1,2])
```
✓ [34] 25ms
```
[[1 2 3]
 [4 5 6]]
2
6
```

```
a = np.random.randint(1,20, size = (3,4))
b = np.random.randint(1,20, size = (3,4))
print(a)
print(b)
print(a@b.T)
```
✓ [35] 10ms
```
[[15 12  4  9]
 [ 3 10 19 18]
 [ 8  4 12 16]]
[[17 18  1  5]
 [18 17  4  7]
 [ 8 18 19 11]]
[[520 553 511]
 [340 426 763]
 [300 372 540]]
```

```
mat = np.random.randint(1,50, size = (3,4))
print(mat)
print(mat.T)
```
✓ [36] 16ms
```
[[ 4 11 21 18]
 [38  9 24 26]
 [28 24 16 44]]
[[ 4 38 28]
 [11  9 24]
 [21 24 16]
 [18 26 44]]
```

```
a = np.random.randint(1, 10, size = 20)
b = np.random.randint(1, 10, size = 20)
print(a)
print(b)
```
✓ [37] 21ms
```
[4 2 1 3 1 1 7 3 4 8 2 1 6 7 7 7 3 2 8 8]
[6 2 3 1 5 7 6 7 8 4 2 7 7 3 2 1 5 8 1 2]
```

```
arr = np.concatenate((a, b))
ar1 = arr.reshape(2, 20)
print(ar1)
```
✓ [38] 16ms
```
[[4 2 1 3 1 1 7 3 4 8 2 1 6 7 7 7 3 2 8 8]
 [6 2 3 1 5 7 6 7 8 4 2 7 7 3 2 1 5 8 1 2]]
```

```
c = np.random.randint(1, 10, size = 20)
d = np.random.randint(1, 10, size = 20)
print(c)
print(d)
```
✓ [39] 25ms
```
[1 9 5 2 8 5 4 2 9 4 9 2 5 6 3 4 6 4 4 6]
[6 4 2 2 5 3 6 8 3 1 3 2 8 6 4 6 3 9 8 4]
```

```
arr = np.concatenate((c, d))
ar2 = arr.reshape(2, 20)
print(ar2)
```
✓ [40] 13ms
```
[[1 9 5 2 8 5 4 2 9 4 9 2 5 6 3 4 6 4 4 6]
 [6 4 2 2 5 3 6 8 3 1 3 2 8 6 4 6 3 9 8 4]]
```

```
print(ar1+ar2)
print(ar1-ar2)
print(ar1@ar2.T)
```
✓ [41] 26ms
```
[[ 5 11  6  5  9  6 11  5 13 12 11  3 11 13 10 11  9  6 12 14]
 [12  6  5  3 10 10 12 15 11  5  5  9 15  9  6  7  8 17  9  6]]
[[ 3 -7 -4  1 -7 -4  3  1 -5  4 -7 -1  1  1  4  3 -3 -2  4  2]
 [ 0 -2  1 -1  0  4  0 -1  5  3 -1  5 -1 -3 -2 -5  2 -1 -7 -2]]
[[395 425]
 [415 429]]
```

```
data = [10,20,30,40]
p = pd.Series(data)
print(p)
```
✓ [42] 13ms
```
0    10
1    20
2    30
3    40
dtype: int64
```

```
d = {"name": ["Aadhil", "Aakash", "Aaron"],
     "Age": [20, 20, 20],
     "Marks": [48, 46, 47],}
df = pd.DataFrame(d, index=["student1", "student2", "student3"])
print("--------------------------")
print(df)
print("--------------------------")
```
✓ [58] 19ms
```
--------------------------
           name  Age  Marks
student1  Aadhil   20     48
student2  Aakash   20     46
student3   Aaron   20     47
--------------------------
```

```
d = {"name": ["Aadhil", "Aakash", "Aaron", "Aayilya", "Abhishekhk", "Abhishekh5", "Abhijit", "Adwaith", "Agnes", "Ahmed"],
     "Age": [20, 20, 20, 20, 20, 20, 20, 20, 20, 20],}
df = pd.DataFrame(d, index=["student1", "student2", "student3", "student4", "student5", "student6", "student7", "student8", "student9",
 "student10"])
print("--------------------------")
print(df)
print("--------------------------")
```
[43]
```
--------------------------
             name  Age
student1   Aadhil   20
student2   Aakash   20
student3    Aaron   20
student4  Aayilya   20
student5  AbhishekhN   20
student6  Abhishekh5   20
student7   Abhijit   20
student8   Adwaith   20
student9    Agnes   20
student10   Ahmed   20
--------------------------
```

```
print(df.head())
```
✓ [44] 19ms
```
             name  Age
student1   Aadhil   20
student2   Aakash   20
student3    Aaron   20
student4  Aayilya   20
student5  AbhishekhN   20
```

```
print(df.head(3))
```
✓ [45] 15ms
```
           name  Age
student1  Aadhil   20
student2  Aakash   20
student3   Aaron   20
```

```
print(df.tail())
```
✓ [46] 16ms
```
             name  Age
student6  Abhishekh5   20
student7   Abhijit   20
student8   Adwaith   20
student9    Agnes   20
student10   Ahmed   20
```

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 10 entries, student1 to student10
Data columns (total 2 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   name    10 non-null     object
 1   Age     10 non-null     int64
dtypes: int64(1), object(1)
memory usage: 240.0+ bytes
None
```

```
print(df.shape)
```

```
(10, 2)
```

```
print(df.loc["student1"])
```

```
name    Aadhil
Age         20
Name: student1, dtype: object
```

```
print(df.iloc[0])
```

```
name    Aadhil
Age         20
Name: student1, dtype: object
```

```
print("------------------------------")
print(df.loc["student2"])
print("------------------------------")
print(df.iloc[2])
print("------------------------------")
```

```
------------------------------
name    Aakash
Age         20
Name: student2, dtype: object
------------------------------
name    Aaron
Age         20
Name: student3, dtype: object
------------------------------
```

```
df["grade"] = ["A+", "A", "A+"]
print(df)
```

```
            name  Age  Marks grade
student1  Aadhil   20     48    A+
student2  Aakash   20     46     A
student3   Aaron   20     47    A+
```

```
df.drop("grade", axis=1, inplace=True)
print(df)
```

```
            name  Age  Marks
student1  Aadhil   20     48
student2  Aakash   20     46
student3   Aaron   20     47
```

```
print(df.isnull())
print(df.isnull().sum())
```

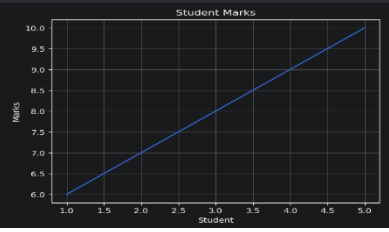
```
            name    Age  Marks
student1  False  False  False
student2  False  False  False
student3  False  False  False
name      0
Age       0
Marks     0
dtype: int64
```

```
df.fillna(0)
df.dropna()
```

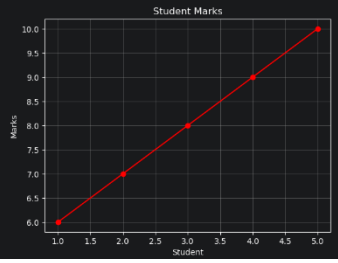
	name	Age	Marks
student1	Aadhil	20	48
student2	Aakash	20	46
student3	Aaron	20	47

```
import matplotlib.pyplot as plt
```

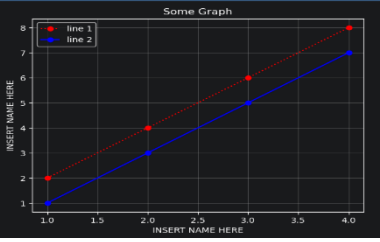
```
X = [1, 2, 3, 4, 5]
Y = [6, 7, 8, 9, 10]
plt.plot(X,Y)
plt.title("Student Marks")
plt.xlabel("Student")
plt.ylabel("Marks")
plt.grid(True)
plt.show()
```




```
X = [1, 2, 3, 4, 5]
Y = [6, 7, 8, 9, 10]
plt.plot(X,Y, marker='o', linestyle=':', color='red')
plt.title("Student Marks")
plt.xlabel("Student")
plt.ylabel("Marks")
plt.grid(True)
plt.show()
```

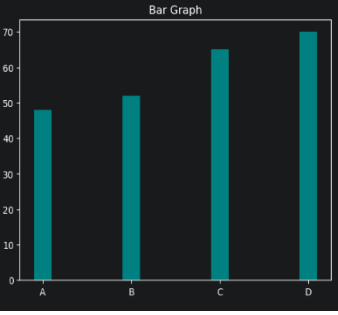



```
X = [1, 2, 3, 4]
Y = [2, 4, 6, 8]
Z = [1, 3, 5, 7]
plt.plot(X,Y, marker='o', linestyle=':', color='red', label="line 1")
plt.plot(X,Z, marker='o', linestyle='-', color='blue', label='line 2')
plt.legend()
plt.title("Some Graph")
plt.xlabel("INSERT NAME HERE")
plt.ylabel("INSERT NAME HERE")
plt.grid(True)
plt.show()
```

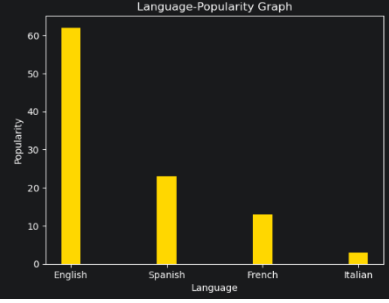



```
A = ['A', 'B', 'C', 'D']
B = [48, 52, 65, 70]
plt.bar(A, B, width = 0.2, color = "teal")
plt.title("Bar Graph")
```

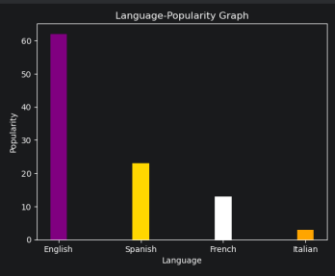
```
Text(0.5, 1.0, 'Bar Graph')
```



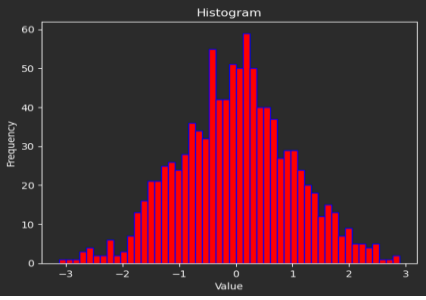

```
L = ["English", "Spanish", "French", "Italian"]
P = [62, 23, 13, 3]
plt.bar(L,P, width = 0.2, color = "gold")
plt.title("Language-Popularity Graph")
plt.xlabel("Language")
plt.ylabel("Popularity")
plt.show()
```



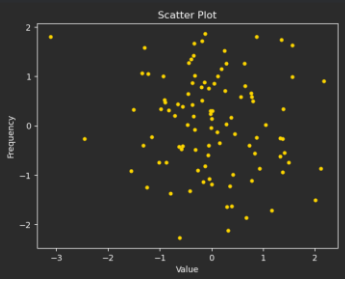
```
L = ["English", "Spanish", "French", "Italian"]
P = [62, 23, 13, 3]
C = ['purple' , 'gold', 'white', 'orange']
plt.bar(L,P, width = 0.2, color = C)
plt.title("Language-Popularity Graph")
plt.xlabel("Language")
plt.ylabel("Popularity");
plt.show()
```
[38] 110ms



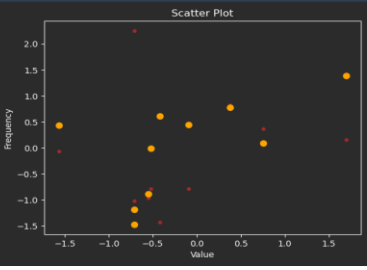

```
data = np.random.randn(1000)
plt.hist(data, bins = 50, color = 'red', edgecolor = 'blue')
plt.title("Histogram")
plt.xlabel("Value");
plt.ylabel("Frequency");
plt.show()
```
[49] 107ms




```
X = np.random.randn(100)
Y = np.random.randn(100)
plt.scatter(X, Y, color = 'gold', marker = '.', s=50)
plt.title("Scatter Plot")
plt.xlabel("Value");
plt.ylabel("Frequency");
plt.show()
```
[57] 124ms

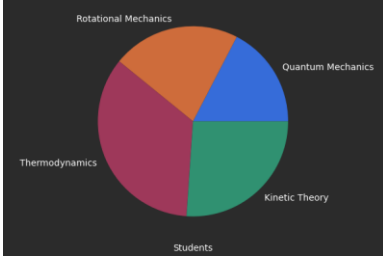



```
X = np.random.randn(10)
Y = np.random.randn(10)
Z = np.random.randn(10)
plt.scatter(X, Y, color = 'brown', marker = '.', s=50)
plt.scatter(X, Z, color = 'orange', marker = 'o', s=50)
plt.title("Scatter Plot")
plt.xlabel("Value");
plt.ylabel("Frequency");
plt.show()
```
[63] 143ms

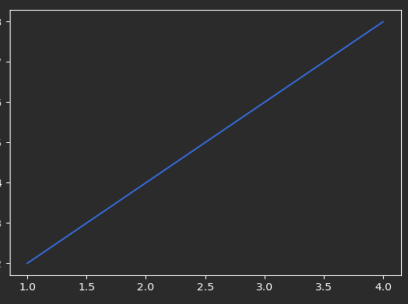



```
Sub = ["Quantum Mechanics", "Rotational Mechanics", "Thermodynamics", "Kinetic Theory"]
Students = [20, 25, 40, 30]
plt.pie(Students, labels = Sub)
plt.title("Pie Plot")
plt.xlabel("Students");
plt.show
```
[66] 104ms

```
<function matplotlib.pyplot.show(close=None, block=None)>
```




```
x = [1, 2, 3, 4]
y = [2, 4, 6, 8]
sns.lineplot(x=x, y=y)
plt.show()
```
[70] 133ms



```
x = [1, 2, 3, 4]
y = [2, 4, 6, 8]
sns.barplot(x=x, y=y)
plt.show()
```
[71] 132ms

